

A
Mini Project Report
on
MedSense: Smart AI symptoms based recommendation solutions
Submitted in partial fulfillment of the requirements for the
degree

Third Year Engineering – Information Technology
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CERTIFICATE

This is to certify that the Mini Project-2B report on **MedSense: Smart AI symptoms based recommendation solutions** has been submitted by **Waseem Khan (22104050)**, **Pushkar Karnik (22104102)**, **Gargi Bora (22104047)** and **Atharva Kavale (22104061)** who are bonafide students of A. P. Shah Institute of Technology, Thane as a partial fulfillment of the requirement for the degree in **Information Technology**, during the second half of academic year **2024-2025** in the satisfactory manner as per the curriculum laid down by University of Mumbai.

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ABSTRACT

The ML-based MedSense: Smart AI Symptom-Based Recommendation Solution is designed to enhance early diagnosis and personalized healthcare by leveraging machine learning algorithms for symptom-based disease prediction. This project provides users with an intuitive Flask-based web application where they can input symptoms and receive instant medical recommendations, including dietary advice, exercise routines, precautions, and suggested medications. By utilizing machine learning models such as Support Vector Classifier, Random Forest, and Gradient Boosting Classifier, the system ensures high accuracy in disease prediction. A structured database efficiently manages medical data, enabling seamless integration with various healthcare applications. This solution plays a crucial role in reducing unnecessary hospital visits by offering initial symptom-based guidance, integrating with pharmaceutical platforms for over-the-counter medication suggestions, and serving as a valuable resource for students, researchers, and AI enthusiasts exploring healthcare applications of machine learning. Additionally, it is highly beneficial for individuals in remote areas, providing preliminary medical recommendations where healthcare services are limited. The project is built on a Python-based technology stack, incorporating Flask, Scikit-learn, Pandas, and NumPy, ensuring a robust, efficient, and scalable system. By integrating modern machine learning techniques with a user-friendly interface, this platform bridges the gap between technology and healthcare, enabling proactive disease management and promoting accessible, data-driven medical assistance.

Chapter 01

Introduction

In today's fast-paced world, timely and accurate healthcare solutions are crucial for improving patient outcomes and reducing the burden on medical professionals. Our AI-powered Smart Healthcare Solution is designed to revolutionize the way individuals access medical assistance by providing instant disease predictions and personalized medical recommendations based on user-inputted symptoms. This intelligent Flask-based web application utilizes advanced machine learning algorithms such as Support Vector Classifier, Random Forest, and Gradient Boosting Classifier to ensure high-accuracy disease prediction. By analysing symptoms entered by users, the system offers tailored medical advice, including recommended medications, necessary precautions, exercises, and dietary guidelines, helping individuals take proactive steps toward better health.

The platform is designed for seamless accessibility, reliability, and efficiency, making healthcare support available anytime, anywhere. Whether used for preliminary symptom evaluation to reduce unnecessary hospital visits, integrated into pharmaceutical applications for over-the-counter (OTC) medication guidance, or deployed in remote areas with limited healthcare facilities, this solution provides a crucial first step in informed medical decision-making.

Backed by a robust technology stack consisting of Python, Flask, Scikit-learn, Pandas, NumPy, and Huggingface, the system ensures scalability, efficiency, and secure data management. Its intuitive user interface makes it accessible to individuals without medical expertise while also serving as a valuable tool for students, researchers, and AI enthusiasts exploring healthcare applications of machine learning. By bridging the gap between AI and healthcare, this smart system empowers users with instant, data-driven medical insights, transforming traditional healthcare accessibility and promoting proactive, informed health management in a digital era.

In addition to providing fast and reliable disease predictions, this AI-driven healthcare solution enhances medical decision-making by offering data-backed insights tailored to each user. The system not only assists individuals in understanding potential health concerns but also helps them take preventive measures through personalized recommendations on medications, lifestyle adjustments, and dietary plans. With its scalable and efficient architecture, the platform ensures seamless performance, making it suitable for individual users, healthcare professionals, and pharmaceutical integrations. Furthermore, the structured database enhances data storage and retrieval, enabling smooth interactions and quick response times. By leveraging machine learning and Flask, this solution aims to bridge the gap between healthcare accessibility and technology, ensuring that quality medical guidance is just a few clicks away—anytime, anywhere.

1.1 Purpose:

This project aims to develop an AI-powered healthcare solution that accurately predicts diseases and provides personalized medical guidance based on user-inputted symptoms. To improve accessibility and relevance, users can initially classify themselves as BP patients, diabetic patients, or normal users, enabling tailored health recommendations. By leveraging machine learning algorithms such as Support Vector Classifier, Random Forest, and Gradient Boosting Classifier, the system delivers high-accuracy predictions, helping users take proactive measures before seeking professional care. This approach enhances telemedicine consultations, reduces unnecessary hospital visits, and supports remote healthcare services.

Built with Python, Flask, Scikit-learn, Pandas, NumPy, and Hugging Face models, the web application ensures a user-friendly and scalable experience. Users can input symptoms and receive instant recommendations on medications, precautions, exercise routines, and diet plans. A structured database ensures efficient data management and quick system response, making it suitable for individual users, healthcare providers, and pharmaceutical applications. Additionally, the project serves as a learning platform for students and researchers, promoting innovation in AI-driven healthcare solutions that are accessible anytime, anywhere.

1.2 Problem Statement:

In today's world, accessing timely and reliable healthcare remains a challenge, especially for individuals in remote areas or those without immediate medical assistance. Delayed diagnosis, unnecessary hospital visits, and the lack of instant symptom analysis often lead to worsening health conditions. Traditional healthcare systems rely heavily on direct doctor consultations, making them time-consuming and sometimes inaccessible. Additionally, pharmaceutical applications lack AI-driven symptom analysis for recommending over-the-counter (OTC) medications. To address these issues, this project develops an AI-powered healthcare system that utilizes machine learning algorithms to predict diseases from user-inputted symptoms and provide personalized medical guidance on medications, precautions, and lifestyle recommendations. By leveraging a Flask-based web application, this solution aims to enhance healthcare accessibility, reduce unnecessary hospital visits, and empower individuals with instant, data-driven health insights—anytime, anywhere.

1.3 Objectives:

- ❖ To develop an ML-based system for accurate disease prediction, enabling early diagnosis and enhancing healthcare accessibility for individuals, pharmaceutical integrations, and remote healthcare.
- ❖ To provide personalized medical recommendations, including medications, precautions, exercises, and dietary advice.
- ❖ To create a user-friendly Flask web app for symptom input and instant medical guidance.
- ❖ To implement a structured database for efficient medical data storage and seamless integration.
- ❖ To develop a system that offers guidance exclusively on primary or basic over-the-counter (OTC) medications, based on user-inputted symptoms, to support informed self-care decisions for common ailments.

1.4 Scope:

- ❖ Can be useful for BP, Diabetic, Normal Patients depending upon their symptoms.
- ❖ Can be used for initial symptom guidance, reducing unnecessary hospital visits.
- ❖ Can be integrated into pharmaceutical apps to suggest OTC medications and precautions.
- ❖ Can be helpful for students, researchers, and AI enthusiasts in healthcare ML applications.
- ❖ Can be deployed in remote areas to provide preliminary medical recommendations.
- ❖ Can be a source for affordable generic OTC medication recommendations.

Chapter 02

Literature Review

- 1) [1] Evaluation of Artificial Intelligence-Based Symptom Checkers by A. O'Connor. This article evaluates the performance of several AI-based symptom checkers, comparing them to medical professionals. It assesses their diagnostic accuracy, highlighting the challenges AI faces in replacing human expertise.
- 2) [2] Deep Learning for Healthcare: A Survey by M. A. Green and S. M. Smith. This paper surveys the use of deep learning methods in healthcare, particularly in predictive diagnosis based on patient symptoms.
- 3) [3] AI in Symptom-Based Diagnostic Decision Support: A Systematic Review by J. O. Ortega and D. M. Rodriguez. This review discusses the use of AI for diagnostic decision support, emphasizing symptom-based systems and their integration into clinical settings. It reviews different approaches to implementing AI systems in healthcare.
- 4) [4] Artificial Intelligence in Healthcare: A Survey of Applications and Challenges in Indian Healthcare Systems by S. Arora, S. Soni. It explores various AI-based systems for symptom-based diagnosis and predictive analytics in medical conditions. The paper discusses challenges such as data privacy, integration of AI with existing healthcare infrastructure, and the potential of AI in addressing the high burden of diseases like diabetes, cancer, and cardiovascular disorders in India.
- 5) [5] Machine Learning Approaches for Disease Prediction and Diagnosis by L. Patel and R. Kumar. This study explores various machine learning techniques used in disease prediction, analyzing their accuracy and efficiency in diagnosing conditions based on symptom patterns. It highlights the role of algorithms like decision trees, random forests, and neural networks in improving diagnostic processes.
- 6) [6] AI-Powered Chatbots in Healthcare: Enhancing Patient Engagement and Diagnosis by K. Thompson and E. Rivera. This paper examines the implementation of AI-driven chatbots in healthcare, particularly in providing symptom-based guidance and preliminary diagnoses. It evaluates the effectiveness of AI chatbots in assisting patients before consulting medical professionals, reducing the burden on healthcare systems.

Chapter 03

Project Design

3.1 Features and Functionality

- ❖ **Disease Prediction:** The system utilizes machine learning algorithms to analyse user-provided symptoms and predict possible diseases with high accuracy. By processing input data efficiently, it assists users in understanding potential health risks and provides preliminary insights before seeking professional medical consultation.
- ❖ **AI Chatbot for Healthcare Assistance:** An interactive AI-powered chatbot is integrated into the system to provide instant responses to health-related queries. The chatbot enhances user experience by offering basic medical advice, suggesting possible conditions based on symptoms, and guiding users toward the next steps in seeking healthcare assistance.
- ❖ **Diet and Nutrition Planning:** The system provides personalized diet and nutrition recommendations based on the user's health condition and symptoms. By analyzing input data, it suggests balanced meal plans, essential nutrients, and dietary adjustments to promote better health. This feature helps users maintain a healthier lifestyle by offering tailored nutrition guidance aligned with their medical needs.
- ❖ **User-Friendly Application:** The platform is built using Flask, ensuring a smooth and intuitive user experience. The web interface is designed with simplicity in mind, allowing users to navigate effortlessly between different features. The structured layout ensures that users can easily input their symptoms, receive results, and interact with the AI chatbot without technical difficulties.
- ❖ **Real-Time Analysis and Insights:** The system processes user data instantly, delivering fast and accurate health assessments. By leveraging machine learning models, the platform provides real-time predictions and insights, enabling users to receive immediate feedback on their symptoms. This functionality enhances the system's efficiency and usability, making it a reliable tool for preliminary health evaluation.
- ❖ **Affordability-Prioritized Recommendations:** This feature is designed to guide users towards cost-effective over-the-counter (OTC) medication choices by prioritizing and highlighting affordable generic alternatives when providing medication recommendations based on user-inputted symptoms.

3.2 System Architecture

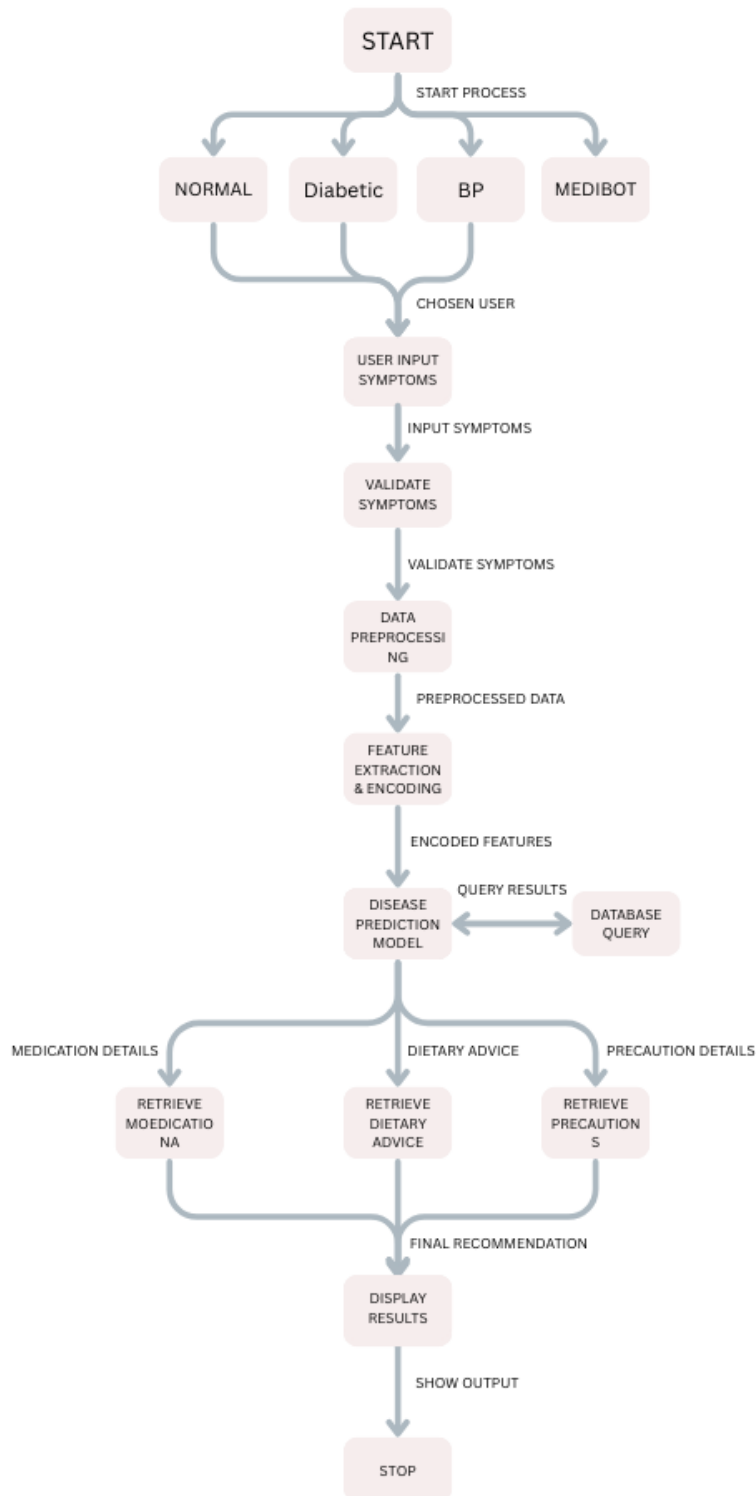


Fig 3.2.1 System Architecture

Fig 3.2.1 depicts MedSense's system architecture which streamlines disease prediction and personalized recommendations. Users choose their category, input symptoms, and undergo feature extraction and encoding. Machine learning algorithms analyse symptoms, retrieve data, and generate tailored results. The system provides medication details, dietary advice, and precautionary measures, forming a final recommendation. The results are displayed to the user, providing a seamless healthcare experience with accuracy and accessibility.

3.3 DFD (Data Flow Diagram)

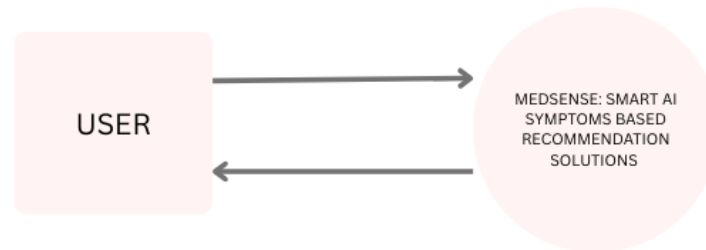


Fig 5.2.1 DFD Level 0

Fig 5.2.1 depicts the Level 0 Data Flow Diagram (DFD) is a high-level representation of an AI-Powered Smart Healthcare System, focusing on the interaction between the user and the system. It focuses on the flow of data, without detailing its internal workings, allowing for deeper decomposition in subsequent levels.

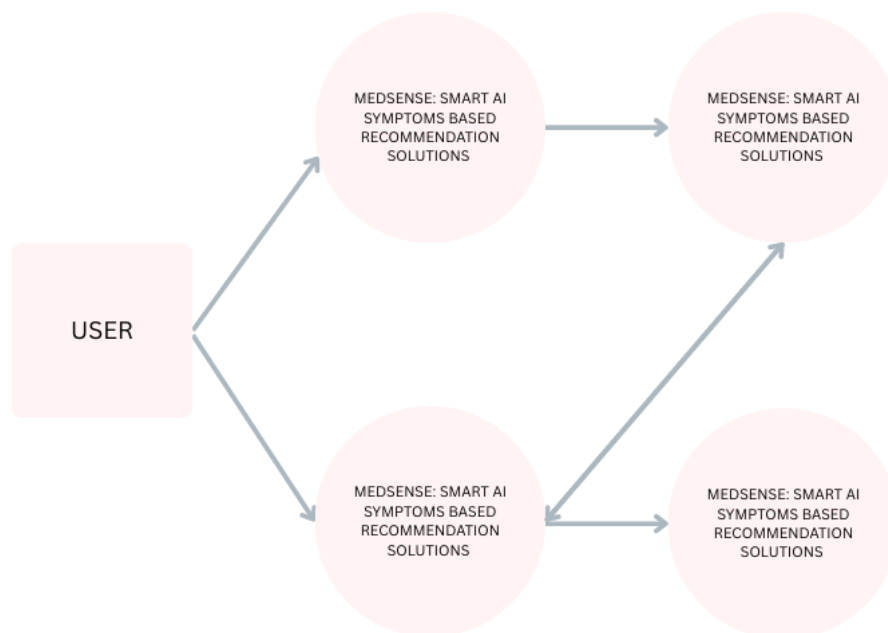


Fig 5.2.2 DFD Level 1

Fig 5.2.2 Illustrates the Level 1 Data Flow Diagram (DFD) breaks down an AI-Powered Smart Healthcare System into key sub-processes, illustrating the flow of data between them. The system starts with the User providing symptom-related data, which is processed by the Disease Prediction Model using advanced machine learning algorithms. It integrates with a database, an AI Chatbot, and personalized medical recommendations. The structured data flow ensures accurate disease predictions, efficient data management, and seamless user interaction, enhancing accessibility and reliability in AI-powered healthcare solutions.

3.4 Use Case Diagram

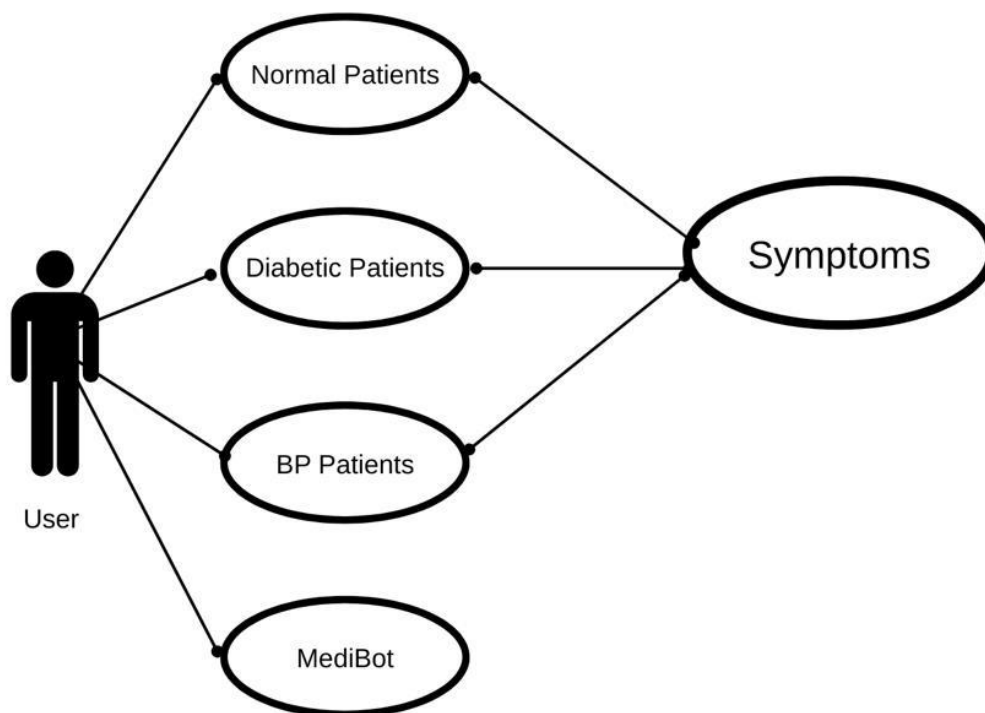


Fig 5.1.1 Use Case Diagram

Fig 5.1.1 Illustrates the use case diagram of MedSense System, categorized into Normal, Diabetic, and BP Patients, and an AI-powered MediBot provide a personalized healthcare experience. Users can input symptoms and receive tailored responses, including predicted diseases, medication details, dietary recommendations, and precautionary measures. The MediBot acts as an interactive assistant, offering AI-driven guidance and answering symptom-related queries. This structured approach ensures accurate medical insights and efficient recommendations.

Chapter 04

Technical Specification

- **Python:** Python serves as the core programming language for the project, offering flexibility, simplicity, and a wide range of libraries for data analysis and machine learning. It facilitates efficient development and implementation of AI-driven healthcare solutions.
- **Flask Framework:** Flask is used as the lightweight web framework to build the platform's backend. It enables seamless communication between the user interface and the machine learning models, handling HTTP requests, routing, and API integration efficiently.
- **Scikit-learn:** Scikit-learn is utilized for implementing machine learning algorithms such as Support Vector Classifier, Random Forest, and Gradient Boosting. It provides robust tools for data preprocessing, model training, and performance evaluation, ensuring accurate disease prediction.
- **NumPy:** NumPy is employed for numerical computations and efficient handling of multi-dimensional arrays. It plays a crucial role in mathematical operations, data transformation, and optimizing the performance of machine learning models.
- **Pandas:** Pandas is used for data manipulation and analysis, allowing structured handling of patient records, symptom datasets, and disease classification inputs. Its powerful data structures ensure efficient data preprocessing and storage.
- **Hugging Face:** Hugging Face's pre-trained models enhance the chatbot's capabilities, enabling advanced natural language processing (NLP) for seamless medical assistance. It helps the chatbot understand and respond to user queries with accurate and context-aware information.

Chapter 05

Project Implementation

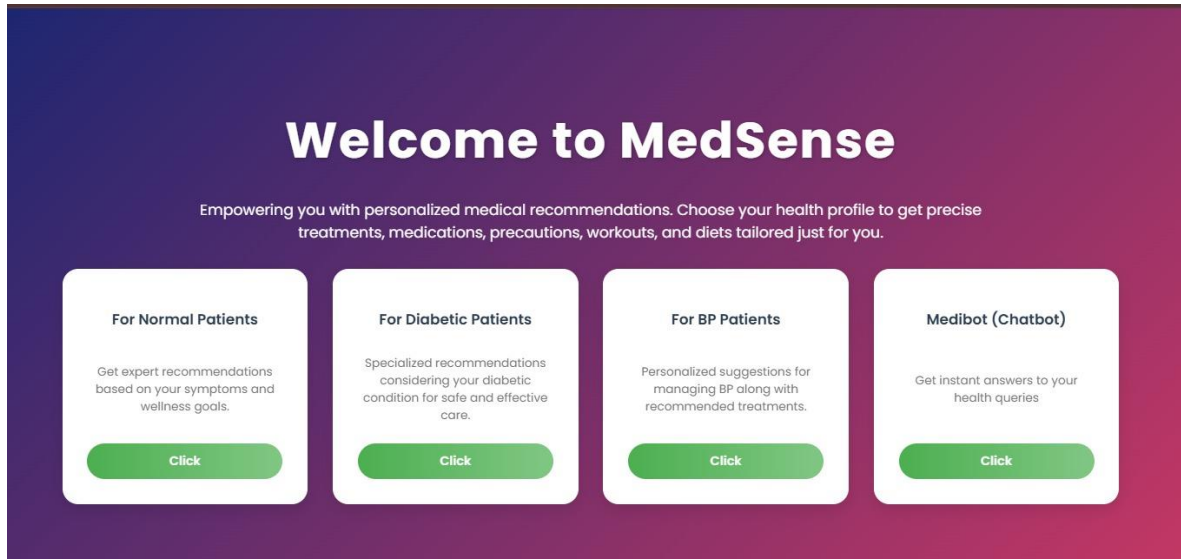


Fig 6.1 Basic input page

Fig 6.1 illustrates the primary page where the patient selects their category, determining the appropriate medication plan. Additionally, a link to the Medical Assistance Chatbot is provided for further support.

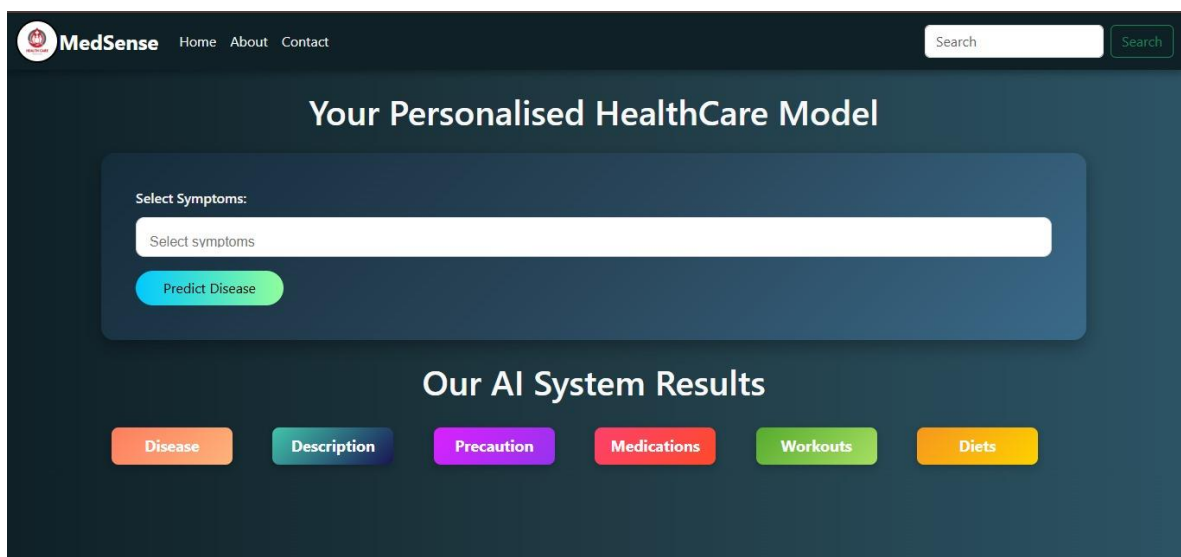


Fig 6.2 Symptoms input from patient

Fig 6.2 illustrates the main symptoms sections which are provided by the patient, based on which the system predicts the disease and generates a detailed output, including its description, precautions, recommended medication, necessary workouts or exercises, and a dietary plan to aid in recovery.

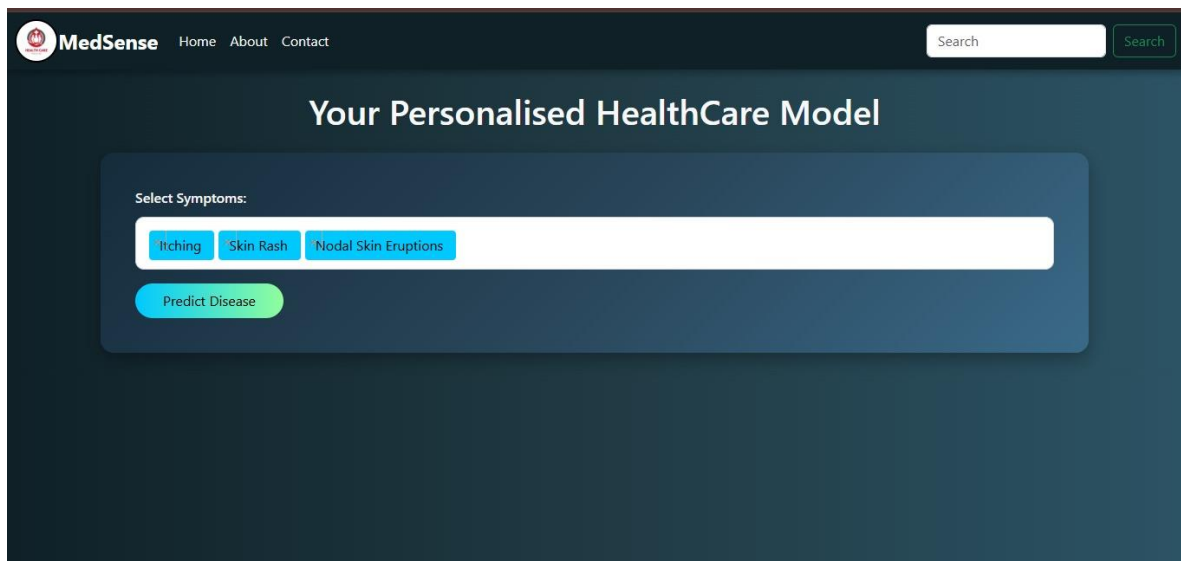


Fig 6.3 Symptoms input from patients

Fig 6.3 depicts the symptom input process, where users can enter multiple symptoms. Upon clicking the "Predict Disease" button, the system analyses the input and displays the diagnosed disease.

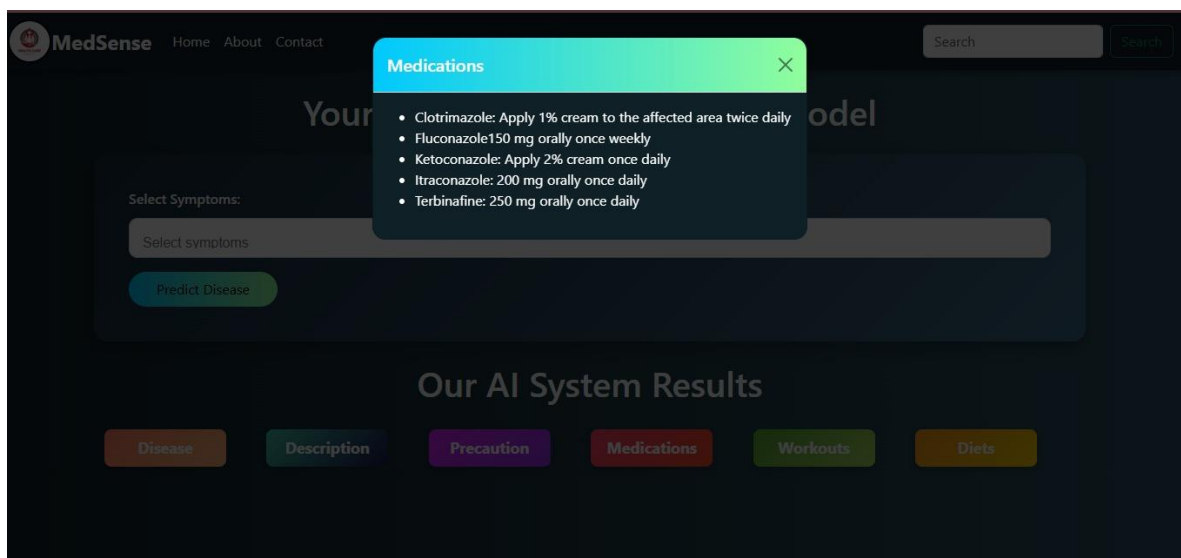


Fig 6.4 Depicts Medications prescribed to patient

Fig 6.4 depicts the medications prescribed by our project for patients, consisting of generic medicines that are cost-effective and affordable while maintaining the same active ingredients as standard medications, ensuring accessibility without compromising effectiveness.

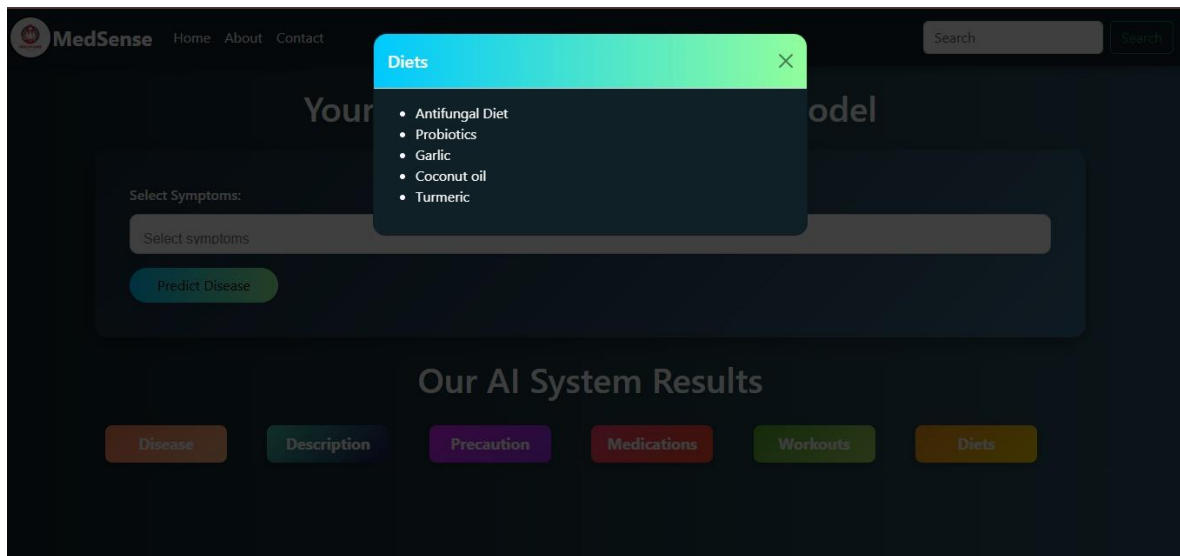


Fig 6.5 Basic diet plan for patient

Fig 6.5 illustrates the diet plan recommended for the patient, tailored to their specific condition to help mitigate the effects of the disease. The diet plan is designed to complement the prescribed medications, ensuring a holistic approach to recovery and overall well-being.

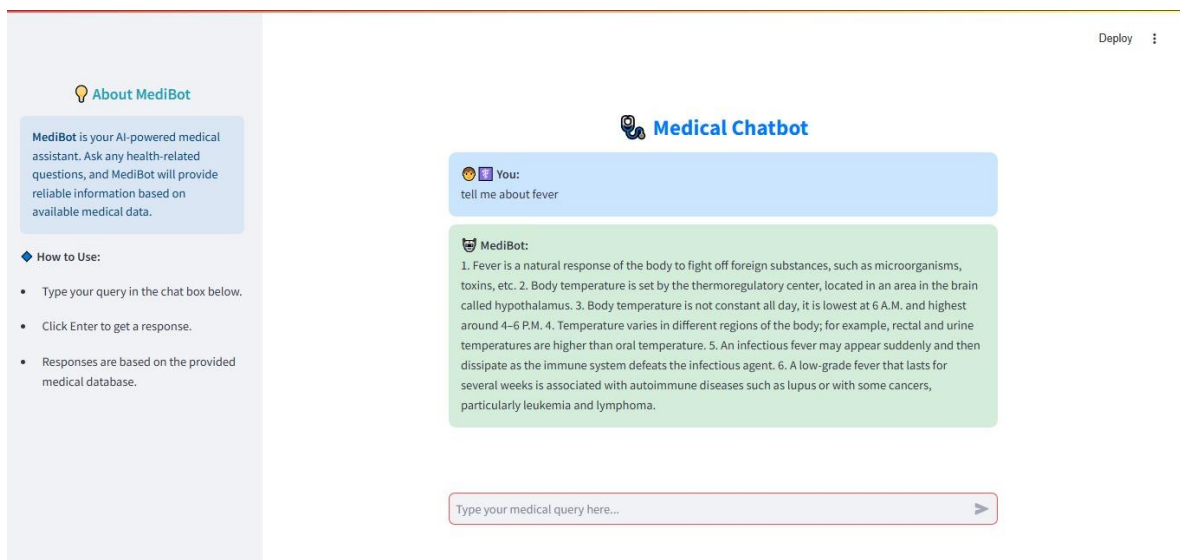


Fig 6.6 Medibot

Fig 6.6 depicts the medical chatbot, Medibot, which serves as a virtual assistant to guide patients by providing proper assistance and resolving their health-related queries anytime and anywhere.

Project Scheduling

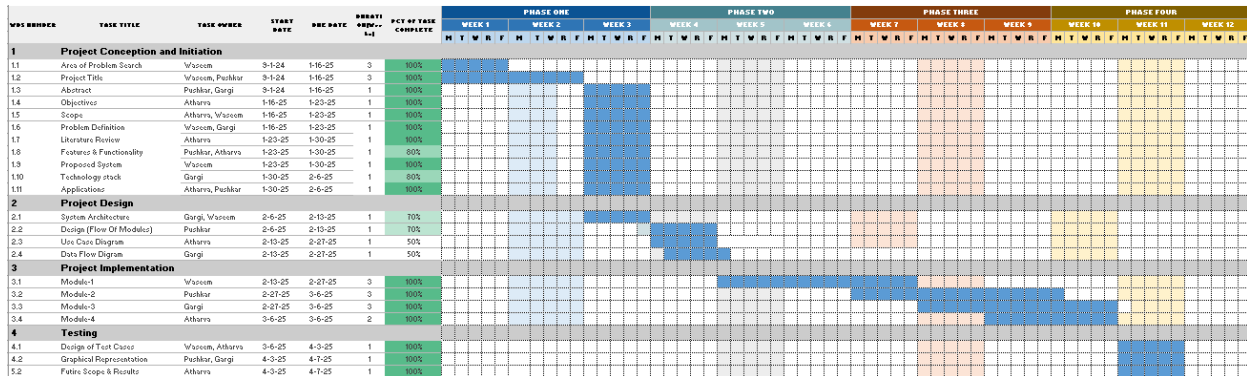


Fig 8.1 Task Allocation and Workflow Distribution

Figure 8.1 Illustrates the project's overall workflow and task distribution among group members, from initiation to completion. It outlines the specific tasks assigned to each member and their respective contributions to the project's development.

Chapter 07

Results

The AI-powered Smart Healthcare Solution has successfully achieved its objective of providing an intelligent, accessible, and efficient platform for disease prediction and preliminary medical guidance. By leveraging machine learning algorithms such as Support Vector Classifier, Random Forest, and Gradient Boosting Classifier, the system delivers accurate predictions based on user-inputted symptoms. This has significantly enhanced the early detection of diseases, allowing users to take timely preventive measures and seek medical attention when necessary.

The system's web-based interface, developed using Flask, ensures ease of use, enabling users to interact with the platform seamlessly. The integration of a structured database has enhanced data management, ensuring quick access to relevant medical information. The intuitive design of the platform, coupled with its ability to provide instant disease predictions, has streamlined the process of preliminary health assessment, reducing the dependency on immediate doctor consultations for minor health concerns.

One of the key achievements of this project is its contribution to improving healthcare accessibility. The AI-driven model empowers individuals, especially those in remote areas with limited access to medical facilities, by providing a reliable, at-home disease prediction tool. This has bridged the gap between individuals and healthcare services, ensuring that users receive timely insights about their health conditions without unnecessary delays.

The project has also demonstrated efficiency in data processing and prediction accuracy. The machine learning models were trained and tested with a diverse dataset, ensuring robustness and reliability in identifying disease patterns based on symptoms. Through rigorous testing, the system has exhibited high precision in disease classification, minimizing errors in prediction and providing users with meaningful health recommendations.

Furthermore, the platform's adaptability and potential for future enhancements make it a scalable solution. With ongoing improvements, such as the integration of real-time health monitoring via IoT devices and telemedicine consultations, the system can evolve into a comprehensive AI-driven healthcare assistant. The success of this project highlights the effectiveness of AI in revolutionizing the healthcare sector, proving that intelligent medical assistance can be both accessible and efficient in promoting early detection, prevention, and personalized health management.

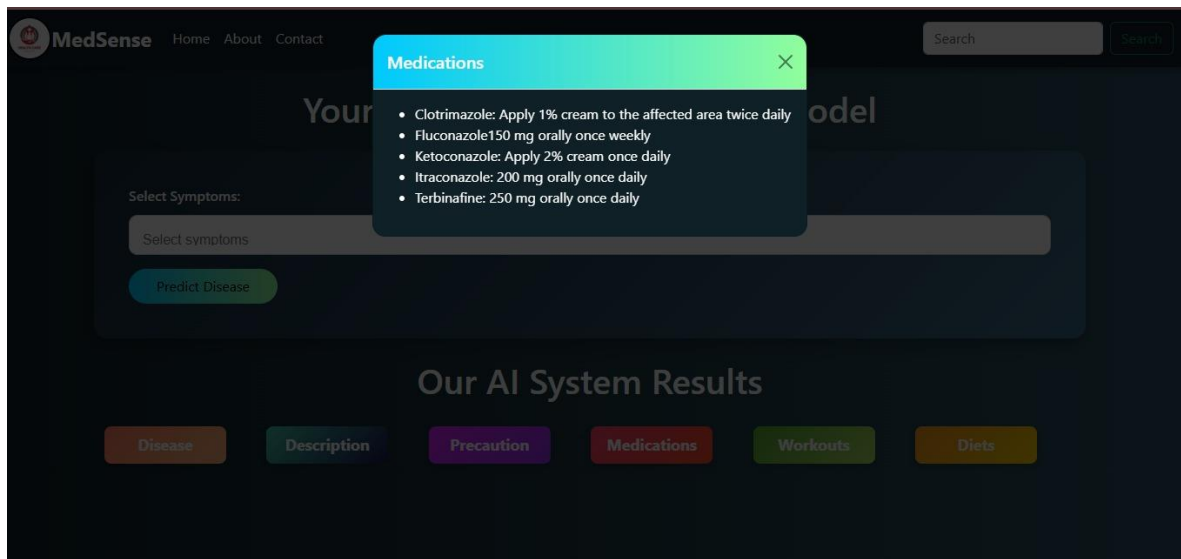


Fig 7.1 Prescribed medications

Fig 7.1 depicts the medications prescribed by our project for patients, consisting of generic medicines that are cost-effective and affordable while maintaining the same active ingredients as standard medications, ensuring accessibility without compromising effectiveness.

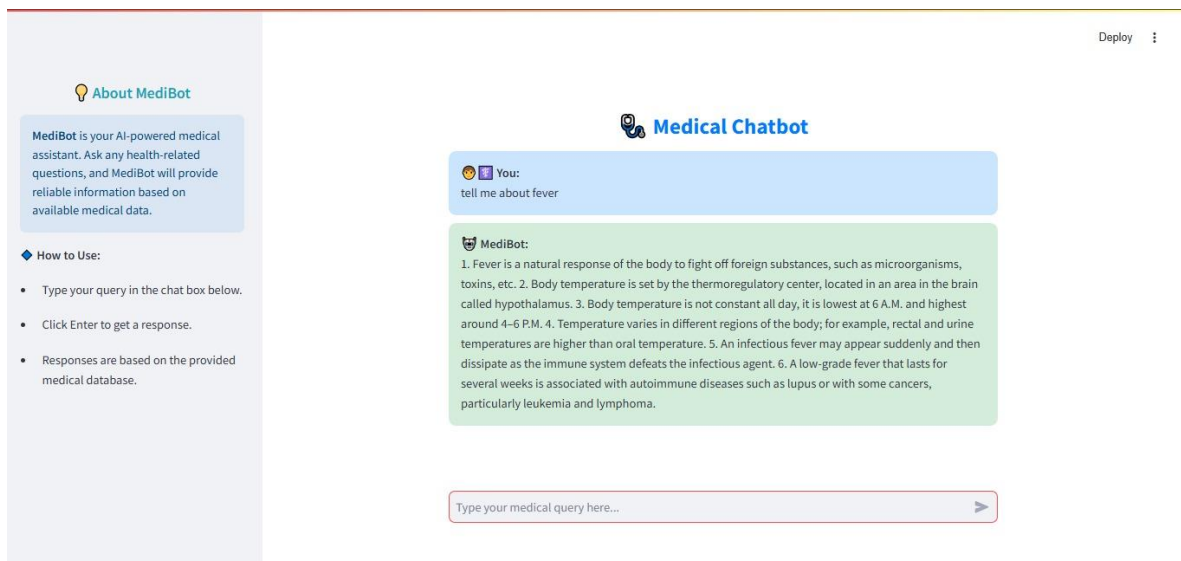


Fig 7.2 MediBot (Chatbot)

Fig 7.2 depicts the medical chatbot, Medibot, which serves as a virtual assistant to guide patients by providing proper assistance and resolving their health-related queries anytime and anywhere.

Chapter 8

Conclusion and Future Scope

Our AI-powered Smart Healthcare Solution presents a revolutionary approach to enhancing healthcare accessibility and disease prediction accuracy. By leveraging advanced machine learning algorithms, including Support Vector Classifier, Random Forest, and Gradient Boosting Classifier, the system provides instant and precise disease predictions based on user-inputted symptoms. The Flask-based web application ensures an intuitive user experience, making it accessible to individuals, healthcare professionals, and researchers. The structured database optimizes data storage and retrieval, ensuring seamless performance. This intelligent system aims to reduce unnecessary hospital visits, support telemedicine services, and integrate with pharmaceutical applications to provide personalized medication recommendations, lifestyle guidance, and precautionary measures, empowering users to take proactive healthcare decisions.

A key strength of this system is its capability to function as a preliminary healthcare assistant, delivering real-time medical insights without immediate doctor consultations. This is especially beneficial for individuals in remote or underserved regions with limited access to medical facilities. The AI-powered disease prediction model continuously learns and evolves, incorporating new medical data to enhance diagnostic accuracy and expand its scope over time.

Looking ahead, the system holds immense potential for further development and expansion. One promising advancement is the integration of Natural Language Processing (NLP), allowing users to describe symptoms in their own words, making the platform more interactive and user-friendly. Additionally, Deep Learning techniques such as Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) could be introduced to improve the system's ability to detect complex and rare diseases with higher precision. Another significant enhancement would be Real-Time Health Monitoring through wearable devices, enabling continuous tracking of vital signs like heart rate, oxygen levels, and blood pressure, providing early alerts for potential health risks.

Moreover, Blockchain technology could be incorporated to enhance data security and ensure safe medical record storage and sharing. This would facilitate seamless integration with hospitals and telemedicine platforms, allowing doctors to access accurate patient histories for better treatment recommendations. Expanding pharmaceutical integrations would enable real-time availability tracking of over-the-counter (OTC) medications, guiding users to nearby pharmacies or facilitating online purchases, benefiting elderly and chronically ill patients. Additionally, multilingual support would make the system more inclusive, allowing users from diverse linguistic backgrounds to access AI-driven medical assistance in their native languages.

By bridging the gap between technology and healthcare, this project empowers users with personalized medical insights, promoting early detection, preventive care, and improved health awareness. As AI and machine learning continue to advance, this solution has the potential to transform modern healthcare, making high-quality medical assistance available anytime, anywhere. With continuous innovation and enhancements, the AI-driven healthcare system will play a crucial role in shaping the future of digital healthcare, ensuring better health outcomes for individuals worldwide.

References

- [1] Rishabh11336, "Medical Diagnosis Flask App," GitHub Repository, [Online]. Available: <https://github.com/rishabh11336/Medical-Diagnosis>.
- [2] Ktrimalrao, "Medicine-Recommendation-System," GitHub Repository, [Online]. Available: <https://github.com/Ktrimalrao/Medicine-Recommendation-System>.
- [3] J. O. Ortega and D. M. Rodriguez, "AI in Symptom-Based Diagnostic Decision Support: A Systematic Review,".
- [4] Abhyast Projects, "Drug Recommendation & Medical Condition Classifier Flask Web App in Python," [Online]. Available: <https://abhyastprojects.com/index.php/product/drug-recommendation-medical-condition-classifier-flask-web-app-in-python/>.
- [5] Siemens Healthineers, "Prediction and Early Identification of Disease Through AI," [Online]. Available: <https://www.siemens-healthineers.com/digital-health-solutions/artificial-intelligence-in-healthcare/ai-to-help-predict-disease>.
- [6] Binariks, "AI/ML Algorithms for Early Disease Detection and Medical Diagnosis," [Online]. Available: <https://binariks.com/blog/ai-machine-learning-for-early-disease-detection/>.